19/04

CPP

What is object orieneted ?
is it programming language?

No, it is a programming Style.

How do we know a language is object oriented programming language or not?
Any language which supports following four features is called object oriented programming languages.

1. Abstraction
2. Encapsulation
3. Polymorphism
4. Inheritance

Abstraction : giving only essential features and hiding implementation details is called Abstraction.

Encapsulation : wrapping or binding of data and functions to be done on them into single entity is called Encapsulation.

Inheritance :deriving a new class from existing class is called Inheritance

Polymorphism : Single name many forms

Any language which is supporting these four features we call them object oriented programming langauges
ex : C++,Java,Python , C# , dotnet

C++ features : BjarneStrouStup.

C++ is a multi paradigm — Support procedural oriented programming principles as well OOPS

C++ is block Structure Programming Language

C++ Supports Generic programming

C++ Supports exceptional handling
C++ has C Library as well as Vast C++ library called STL

Applications of C++

1. video gaming
2. developing OS
3. developing Compilers and Interpreters
4. used in desiging Webdevelopers.
5. For many new langauges it is acting as foundation

.cpp
ex : file.cpp

```
c++ file.cpp
g++ file.cpp
a.out
```

```
printf("hello");
printf("%d",var);
printf("%d %s",v,s);
printf("val is %d",v);
```

cout<< → predefined object of C++ → insertion operator

```
cout<<"hello";
cout<<var;
cout<<v<<s;
cout<<"val is :"<<v;
```

predefined object

cin>>

extraction operator

```
cout<<v<<"  "<<s<<"\n";
```

```
scanf("%d",&var);
scanf("%d  %s  %f",&v,s,&d);
```

```
cin>>var;
cin>>v>>s>>f;
```